JÖNKÖPING UNIVERSITY
School of Education and Communication

Teaching methods for science subjects in elementary schools

A qualitative study on the methods science teachers describe that they use to teach science subjects in elementary schools and what effect they say these methods have on students' motivation towards science subjects.

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ABSTRACT

Rudolph Shinda Ndurya Pages: 37

Teaching methods for science subjects in elementary schools

A qualitative study on the methods teachers describe that they use to teach science subjects in elementary schools

The objective of the empirical study was to explore which teaching methods science teachers in elementary schools describe that they use and the effects these methods have on student motivation towards the subject, according to these teachers.

The study is based on the socio constructivism perspective while also correlating pragmatism and cognitive constructivism in relation to teaching methods in science lessons. Semi-structured interviews have been conducted with experienced and skilled elementary science teachers from around the district of Jönköping. Through the interviews the teachers gave accounts and descriptions of teaching choices they make for their lessons. They also accounted for the tangible effects these methods had on students' motivation in the subject.

Results from the study gave insight into methods science teachers use in their lessons and the reasoning behind their choices. The results leave little room for doubt that teachers work and promote understanding in their lessons. They consider students' interests, experiences, and values when teaching. By using integrated working methods that gave tangible results, like classroom discussions, students' pre-knowledge and experiences, repetition, web assisted learning, investigative working methods and out-door learning, the teachers created a conducive environment for stimulation and greater academic commitment among students.

The results showed that having high expectations for students to do well is only going to work if teachers are also supportive of their students, which, according to the study, was evident in different teaching scenarios. This study revealed that the interviewed teachers have accumulated a repertoire of subject know-how and were actively using it in various ways to impact student learning and motivation.

As teachers then, the right attitude would be to understand which methods fit the class dynamic as this will affect student engagement and subsequently, performance and motivation. It is clear therefore, that the need to adapt teaching to cater for students' individual needs should be deemed a good reason to help unlock students' learning potential in their quest for knowledge acquisition.

Search words: teaching methods in science, students' motivation, science lessons, teaching science in elementary schools.

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“What teachers need to have at their fingertips are strategies for handling children’s questions and turning them to the advantage of investigative learning. They also need sources of information and a level of general understanding that facilitates quick and effective use of these sources.” Harlen (1997).

To,

Amy, Deyzhah, Naeomi, Nenne & Nala.

Thank you for your support and patience throughout this program. You have been my best cheerleaders.

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1. INTRODUCTION

Literature study by Ndurya (2018) indicates teacher's importance and ability to motivate students as being extremely important for the students' continued development in school and in society. During my previous VFU (teaching practice) I came into contact with science teachers who desired to create interest and motivation for science subjects. This interest in working to have students motivated in science subjects led to the 2018 literature studies; "*What methods can science teachers use in their teaching to motivate students in grades 4-6?*" Good teachers help students activate their existing experiences, prior knowledge, and conceptions and help them to build knowledge from these. They also encourage students to practice and apply knowledge on different concepts in different contexts through their teaching methods. For students to have meaning in their science learning, they need help to adequately link what they learn to their existing experiences and conceptions. This is because their understanding of the phenomena in their world might differ from the teachers'.

Earlier research by Strong, Harvey & Robinson (1995), sought to find answers whereby both teachers and students were asked two simple questions; what kind of work do you find totally engaging? And what kind of work do you hate to do? Engaging work came out as a key answer by the students who described it as work that stimulated their curiosity and permitted them to express their creativity, fostering positive relationships with others. Research has also pointed out that teaching is often too focused on textbooks and individual activities and tasks instead of learning in itself (Ewald, 2007).

To counter this, the school, according to the curriculum, has a responsibility to stimulate student creativity, curiosity and self-confidence as well as their willingness to try to translate ideas and solve problems. Students should be given the opportunity to take initiative and responsibility and develop their ability to work independently as well as with others. The school should also promote learning where the individual is stimulated to acquire and develop knowledge and values (Skolverket, 2019). The teacher's ability to approach science teaching in a way that meets students' uniqueness and complexity is therefore extremely important for the students' continued development both as an active individual in school and in society. Therefore, based on the information described above, it is interesting to investigate how science teachers in middle schools describe the reasons behind the methods they use to teach science and what effects these methods have on student motivation towards the science subjects in general.

2. BACKGROUND

2.1 Curriculum

According to the Education Act, education in Swedish schools must be based on scientific research and proven experience, (1-chapter, section 5§ School Act (SFS 2010: 800). The syllabus for the subjects of natural sciences states that through teaching, pupils should be given the opportunity to ask and explore questions about nature and humanity based on their own experiences and current events (Skolverket, 2018). Furthermore, teaching should provide pupils with the opportunities to look for answers to questions by means of practical systematic studies such as practical investigative work to develop their skills (Skolverket, 2018). This implies that teacher's competence, ability, and commitment are of great importance to succeed with the school's mission.

In the Swedish school system, the subjects of biology, physics and chemistry are included in what is referred to as NO (science-based subjects). Knowledge of biology is of great importance for society in such diverse areas as health, natural resource use, and the environment (Skolverket, 2018). Knowledge of physics is important for society in areas such as energy supply, medical treatment, and meteorology (ibid.). This knowledge provides the individual with the tools to be able to contribute to sustainable development in society (Skolverket, 2018). Knowledge in chemistry is vital for health, economy in use of resources, and the development of materials and environmental technology (ibid.). The implication, therefore is that students' education has a wider meaning and significance where, among other things, teachers should help students acquire knowledge that prepares them for living, in work and also the ability to participate in the democracy of the society (Skolverket, 2017).

2.2 Learning Theories

Different learning theories explain what learning and knowledge imply. The study of how we humans learn focuses on how we acquire and modify our knowledge, skills, strategies, beliefs, and behaviors (Schunk, 2012). Learning therefore, involves an enduring change in behavior or in the capacity to behave in a given situation which results from practice or other experiences (ibid.). We possess knowledge through intellectual thoughts and through dialogues and discussions with others. For this qualitative study three learning theories will be summarized. These theories are well established and are a vital part of the education and learning in Sweden. The theories discussed need not necessarily be considered as contradictions to each other as much as complementing the other (ibid.). This is because they relate to different aspects of learning and knowledge acquisition.

2.2.1 Socio-cultural perspective (Lev Vygotsky)

Students develop learn in interaction with others around them. This was the perspective behind Vygotsky's theory which emphasized the importance of social interaction and instruction (Smidt, 2010). Blake and Pope (2008) note that Vygotsky's theory proposes that development does not precede socialization. This connotates that an individual's mental development is centrally dependent on social structures and social relations (ibid.). This coincides with Smidt's (2010) view in the matter, that a socio-cultural perspective on teaching has its background in Vygotsky's theory of learning which delineates that the individual's learning process through communication with others is a prominent aspect of development. It is here the author says, where the interaction takes place in a community. According to Vygotsky, the communicative processes are central as tools for the individual's development (Säljö, 2015). These processes include language, memory and concept formation and arise partly through interaction with other people and partly by the individual mentally processing them. An important part of learning, according to Vygotsky, is that it occurs through interaction between people and through various forms of communication, both linguistic and non-linguistic (Säljö, 2015).

Säljö (2015) goes on to describe that knowledge comes into play and is acquired through interaction between people when they coordinate their perspective and work collectively, hence the origin of knowledge is in the interaction. Subsequently, this implies that knowledge should be understood as something that comes to life and gains meaning in the inter-action between humans (ibid..). Learning isn't merely focused on acquiring skills and knowledge alone, but also on individuals interacting and learning through conversations and discussions with others (ibid.). This becomes a platform for the individual to further develop their everyday experiences. The concept of cognitive learning zone is in play here. According to Vygotsky, the Zone of Proximal Development (ZPD), requires adults or peers to aid students who cannot complete the assignments without help. The ZPD is therefore the gap between what a student is capable of doing independently and what the student requires help to be able to accomplish. Instruction and learning occur in the ZPD because in this zone the students interact and coordinate their perspectives with those who already have the knowledge. This way the students in this zone can be successful with instructional help (Blake & Pope, 2008). Instruction and help come from the teacher who in this perspective is the more competent person giving physical and intellectual support to the individual during their learning (Säljö, 2015). When this happens, Vygotsky believes that teachers have an important role in students' exploration and meaning-making (Smidt, 2010).

The Swedish curriculum states that students should be given the opportunity to take initiative and responsibility and develop their ability to work independently as well as with others (Skolverket, 2018, p. 9). It injects respect for students as well as hope for teachers in their profession, to be able to motivate and guide the students in their knowledge growth. When looking through the glasses of the sociocultural perspective, it is vital for a teacher to have the ability to fathom that learning is context situated. This is when learning is seen as a social process embedded within culture, contexts and activities (Säljö, 2015). When teachers understand the individual's actual experience, that is, the individual's actual level of development, then it is easier to guide them to better understanding. A critical aspect of the development zone involves how a teacher interacts pedagogically with students, where the insightful teacher takes the time to help students in a simple and joyful way, to perform the various tasks of the subjects (Chaiklin, 2003). This is something that can be linked back to the Swedish curriculum where teachers must ensure that every student has the right to develop, feel the growing joy of knowledge, and experience the satisfaction given by making progress and overcoming difficulties (Skolverket, 2018. s 11).

2.2.2 Pragmatism (John Dewey)

John Dewey believed that education should be adapted to meet every child's unique abilities, and that children should be provided with necessary tools and skills to promote their individual growth and development (Erwin & Kipness, 1997). Dewey acknowledged the need for children to be active participants in surroundings that stimulate their imagination, exploration, and interpersonal interactions (ibid.). According to Dewey According to Dewey (cited in Imsen, 2006), individuals learn from experiences they gather from doing things, not by being influenced by external stimulation. Dewey felt that experience is an interaction between doing something and seeing what the ation leads to. The individual learns when he or she understands the connection between what the person did and what the result is. In this way, learning becomes something that one contributes to oneself through activity and action (ibid.). Experiential learning is a process of meaning-making that is vital for a child right from early education. According to Dewey, learning through play and active exploration of our surroundings facilitates progressive knowledge construction within specific social, cultural, and ecological contexts (Luff, 2018). Learning by doing is thus a concept within pedagogy describing that learning can be deepened through various elements of skills training and practical application tasks.

Dewey's approach to learning can be interpreted as that we can learn something new through our own experiences because our thoughts and actions form an entirety to our surroundings (Imsen, 2006). For example, students visiting a zoo to see and learn about animals instead of just reading

about them in a book at school. In this way, the individual can reflect on their experiences. Therefore, Luff (2018) states that teachers in their work observe children's interests and aptitudes in order to understand their past and current experience and to provide worthwhile new experiences. This corresponds to Dewey's concept of aesthetic experience as something enriching and fulfilling, with characteristics of vitality and heightened awareness (Luff, 2018). By supporting children's undergoing of activities, teachers can thus add breadth and depth to their everyday experiences. In doing so, children will benefit by being able to integrate their new experiences with existing knowledge thereby building new insights and understandings. This in turn seems to agree with the Swedish school curriculum in stating that the school should promote students' harmonious development through variation and balanced composition of learning content and working methods (Skolverket, 2017). From Dewey's way of thinking, it can be seen that the practical as well as the intellectual are interconnected and teachers should take this into account in their teaching. Since teaching science is characterized by practical work, learning by doing should be a good way to reach the individual to help them gain experiences that spark their intellectual interest and respect for their environment.

2.2.3 Cognitive development (Jean Piaget)

Blake and Pope (2008) describe that Cognitive psychology is a branch of psychology that focuses on studies of mental processes which include how people think, perceive, remember, and learn. Its core focus is on how people acquire, process, and store information. This coincides with Jean Piaget's fundamental orientation in what is widely accepted that higher cognitive functions in an individual unfold concomitantly as they develop and mature with age. What this means then, is that children's capacity to learn expands and increases as they also grow through the years and experience the world around them. According to Piaget, a child is capable of higher mental functions around them the older the child gets. This explains Piaget's four stages of an individual's cognitive development. Piaget's four stages in cognitive development include sensori-motor, pre-operational, concrete, and formal (Blake & Pope, 2008) In the sensori-motor stage, while in infancy, a child learns by using their five senses, object permanence, and actions that are goal-directed. Children are egocentric because they don't understand that someone else's perspective might differ from their own. This means that they will fail to coordinate their point of view with others (ibid.). The next stage, preoperational stage has a spread from ages two through seven, during which children are able to do one-step logic problems, develop language, continue to be egocentric, and complete operations. However, individuals in this stage do struggle with centering and conservation. The concrete stage has a span of ages seven through eleven after which comes the formal operations

stage. This stage in a child's development spans from age twelve to adulthood. From here on the child can think logically and show lingering egocentrism (ibid.).

As Blake and Pope (2008) note, Piaget's theory implies that development occurs in stages in a child's life as they mature and without help from outside. Piaget's fundamental perspective is thus grounded mainly on the individual learner and not depending upon any outside social stimuli or for that matter inner resources as it is with Vygotsky. Piaget saw the child as an active, intelligent and creative producer of his own knowledge structures in life. Blake & Pope (2008) note that by using Piaget's theory in the classroom, "teachers and students benefit in several ways. Teachers develop a better understanding of their students' thinking" (Blake & Pope, 2008, p.62). This enables them to adapt their teaching methods and strategies to their students' cognitive level in many ways, for example motivational set, modeling, and assignments (Blake & Pope, 2008). The implications for the teacher are that it is of little importance what they know about the student's learning process and expectations, unless they have knowledge to determine when the student is ready to approach a given concept. Kirsh (2014) emphasizes that this way of thinking leaves teachers with a dilemma of deducing if a student is capable of learning a particular concept. Kirsh (2014) goes on to describe that more often than not, the result is that the teacher finds that students are not capable of approaching given concepts because they are "too abstract." Again, we end up in familiar waters where the thinking is that "development precedes learning and that it is a process that is intrinsic to the individual and cannot be significantly influenced by others." (Kirsh, 2014, p. 244).

2.3 Earlier Research

Research has highlighted how paramount it is to balance and interact teaching methods with different learning environments in order to offer students optimal ways to learn. Akram, Norman and Mahmood (2015) identified several key factors for motivation in science learning, one of them being students' knowledge of their performance goals. In other words, students being aware of what is being expected of them to enable them to meet their curriculum goals (ibid.,). Boström (2004) argued likewise that when students are aware of how best they can learn and understand (which also requires that the teacher understands how each individual student learns best), they can take greater responsibility for their learning. This facilitates the use of different teaching methods at different levels in the classroom. Luff (2018) argued for experiential learning where critical and creative thinking and experiences in nature are facilitated. Accordingly, Sjøberg (2010) indicated that it was important for teachers to know students' interests and use them in their teaching, while on the other hand cautioning that teachers cannot plan their teaching based only on students' interests. This is consistent with Hattie (2014) who describes the importance of taking advantage of students'

interests, noting that coupling that with what is learned in science could arouse a curiosity for knowledge and a thirst to learn more. Sjøberg (2005) explicates thus;

> How should teaching material be presented in a way that promotes students' learning? Because it is learning that is the goal, and what a student learns, is not always the same as what the teacher teaches, or what is presented in a textbook (p. 33).

The use of digitalized teaching methods has been shown to increase student motivation and interest in learning in general. According to Hylén (2013) increased motivation for schoolwork and student engagement in learning was achieved with the introduction of digitalized teaching methods when used in combination with other methods. Uluyol and Sahin (2016) concluded in their study that digital tools in teaching science played an important role in student motivation. Teachers accounted for an increase in students' knowledge and skills, such as communication skills, problem solving, self-confidence, understanding, analysis, and expression (ibid.). Earlier research annotates that interactive white boards contribute to motivation and engagement of students to explore the role of science in understanding their outside world and important scientific concepts (Murcia & Sheffield 2010). Existent are also barriers in using digital tools in teaching science. Teaching is evolving and so is the technology being used in schools. Being congruous with the evolution of technology is important as problems do arise when using digital tools in teaching (Chandra & Briskey, 2012). This scenario could potentially cause a loss of teaching time for the students.

Holubova (2008) saw a lot of good in having project-based teaching in science lessons as the benefits were tangible. Apart from the fact that the activities could be done out of school they facilitated for teamwork and social growth. Students also learned to work with the discipline of researchers, among other things (ibid.). Millar (2010) noted that successful practical activities depended on how the teacher presented them to the students in their science lessons as this could have significant consequences on whether the lesson aims and purposes are achieved. According to Millar (2010) practical work in science led to better learning as students understood and remembered what they did themselves.

Earlier research sheds light on outdoor learning activities by describing that students are offered learning in firsthand, reality-based contexts and this could facilitate deeper understanding of subject content (Blom & Fägerstam, 2013). Outdoor learning also facilitated for long-lasting effects on students' cognitive abilities in that students experienced a richer contextualized, hands-on science education as active participants (ibid.). Another benefit was that a student's shared experiences functioned as a foundation for continued learning in the classroom. Hofstein and Rosenfeld (1996)

described that outdoor activities were some of the most valuable informal science learning methods for students. In comparison with classroom learning, outdoor learning activities impacted students' understanding of science (ibid.). This was because learning environments allowed students to interact "physically and intellectually with instructional materials through hands-on experimentation and minds-on reflection" making huge impacts on student learning of science (ibid.). Respectively then, coupling outdoor and indoor learning activities helped diversify the repertoire of learning opportunities among students (Hofstein & Rosenfeld, 1996).

3. AIMS AND RESEARCH QUESTIONS

The purpose of this study is to investigate how teachers in the middle school teach science subjects and which methods they use when teaching their subject. Which effect do these subjects have on student's motivation? In view of this purpose, I would like to examine the following research question:

- Which teaching methods do science teachers in the middle school describe that they use and which effects do these methods have on students' motivation towards the subject, according to these teachers?

4. METHOD

This section contains descriptions of how the empirical study has been conducted, processed, and then analyzed. The choice of method, analysis of the data collection, collection of the empirical material, and ethical considerations have also been accounted for. One of the main aims of interviews in qualitative studies is to understand phenomena from the subject's own perspective in life (Kvale & Brinkmann, 2014). Coming from this background then, it only reasonable to try and comprehend the contextual complexity in the phenomena that are central to the teaching methods of the science teachers. This is through the use of semi structured interviews based on thematic questionnaires (Kvale & Brinkmann, 2014).

4.1 Choice of study method

Semi-structured interviews have been conducted with experienced and skilled elementary science teachers from around the district of Jönköping. Among the advantages of interviews is that the interviewee's voice pitch, mimicry, or pauses do tend to provide information that would otherwise never have been noticed if it was instead written down by the interviewees (Bell, 2006). Interviews are, however, time consuming. One of the reasons I chose this interview method was that I preferred meeting the teacher I am interviewing in person. An important aspect in the study was that there should be no scientific objectivity, instead I wanted to focus on what the science teachers described as their methods of choice in teaching science subjects and their reasons behind the choices. The interviewees got freedom to create their own thoughts into answers. I was, however, flexible when it came to how the interviewees appeared to understand the interview-questions and how they interpreted them.

Stukát (2011), notes that in semi-structured interviews the interviewer prepares a set of questions that are the same for all the interviewees. In addition to that, the interviewer can then follow up on the answers he gets individually. Therefore, I varied my questions in accordance with how the interviewees answered them. I felt this was the best way to go about the interviews as the follow up questions gave me a deeper understanding of the interviewee's descriptions. This agrees with what Bryman (2008) duly explains that semi-structured interviews are based on predetermined questions that the interviewer furthermore can, if needed, follow-up on in order to get clarification.

4.2 Choice of study groups

I have chosen five science teachers to do the interviews with because they have the necessary experiences and knowledge of the phenomena that is being studied here. The study was conducted

by interviewing science teachers who are active in middle school classes. Three of the teachers have a license to teach both in grades 4 to 6 and 7 in Sweden.

I had the aim to interview science teachers active in grades four to six as this would be relevant to the study (Bryman, 2008). I set out to interview eight science teachers, however three of them opted not to be part of the study. Of the five science teachers, three are native Swedish and have been teaching for over 15 years. Two science teachers have, in addition to experience in teaching in Sweden for over 6 years, international teaching backgrounds with several years of experience in teaching in middle school and secondary school both in the U.S.A and U.K. One science teacher was active in grade four this year, and will move with the class to grade five. The remaining two science teachers are active in grade six and seven. I have had the pleasure of working with these teachers in different times during my five years at Jönköping University.

Correspondence with these teachers began by emailing them a letter of consent where I informed them about the study (see Appendix 1). I was subjective in my choices as I already had knowledge of the teachers and knew they had tremendous solid knowledge and experience in science subjects. This gave me a sense of confidence that these science teachers' interviews would fulfill the purpose of this study in the best way possible as they possess both experience and solid knowledge in the field. Denscombe (2016) states that selection is about conscious choices based on what they bring into the study or that there is a knowledge to share.

4.3 Approach

After preparing my interview questionnaire (see Appendix 2) I contacted the teachers by email and by phone. I wanted as much as possible to have open questionnaires to encourage the interviewees to be free to express themselves as they wished. In order to get fulfilling and qualitative answers, I prepared several follow-up questions as I anticipated probable answers to my questions. According to Eklund (2012) it is important that the questions can be adapted depending on interviewees. For my study therefore, I tried to adapt to how the teachers answered and understood the questions in order to encompass the study's purpose. The current worldwide Coronavirus pandemic has among other things, forced social distancing. As a result, the interviews were held through online platforms. The teachers owned the terms and were to decide upon the time and conditions for their interview. This agrees with what Eklund (2012) suggests in that the place where the interview takes place is of importance and that those being interviewed feel comfortable.

The interviewees were assured about confidentiality. After that the purpose of the study was briefly mentioned, with which all were happy to take part (Eklund 2012). All my interviews were recorded using an audio recording application in my mobile phone and also using my laptop. This is for safety purposes, in case something happened to any of the electronic devices, there would be a backup. Among the many positives to recording, the biggest positive for me as the interviewer, was that I got to focus on my subject, giving them the necessary time to think about the questions before answering them. Moreover, recording spares me the task of having to remember every single detail mentioned during the interview as I jot them down (Denscombe, 2016). It took about three weeks to get all the interviews done. As a bonus during the interviews I jotted down things perceived to be important along with where in the interview these important things were said. For example, at 10:40 subject talks about their facial expressions and things they say that students see and know.

I was optimistic when explaining to the interviewees that the interviews would take 30 minutes as only the first interview took almost about that time, at 38 minutes. This may be because the teacher was very precise with their answers, and so there was no direct need for follow-up questions. The rest of the interviews varied in times and took up to 60 minutes. This was due to the fact that I found it necessary to elaborate on the questions, while the teachers' interesting answers also created room for follow-up questions.

4.4 Research ethical aspects

Considering research ethical aspects, I gave my word that the materials I got from the interviews would be solely used for this study. This was to ensure interviewees' confidentiality in accordance with Kvale & Brinkman (2014). Before the interview started, I guaranteed the interviewees that their participation was voluntary and that they had the right to cancel the interview at any time they wanted. Vetenskapsrådet (2017) describes four principles applicable to humanities social science research that I used as reference.

The information requirement principle – As the one doing this research, I had to inform the interviewees about their role and conditions of participation in the study. I let them know that participation was voluntary and could be cancelled anytime. Among the information shared was the aim of the study.

The consent requirement principle – The teachers gave consent to be participants in the study I was conducting. They freely chose to be interviewed and recorded by me.

Confidentiality requirement principle – The participants in this study were notified that the use of their information would be confidentially treated and that personal data would be kept away from unauthorized persons. In other words, the information would not be traced back to them in any way.

Usage requirement – Information the teachers gave me was only going to be used for research purposes. This also means that the teachers' identity will be secured in the preparation of this study. In my study I will use codes to depict the teachers in order to meet the criterium of the confidentiality principle: T 1, T 2, T 3, T 4, and T 5.

4.5 Trustworthiness and Transparency

Trustworthiness of this empirical study has been achieved through both confidentially collected and methodically interpreted and analyzed data to ensure its quality. A concise transcription process of interview data with citations relayed in the study as they were uttered by individual teachers was relevant in order to maintain dependability and reader confidence of the qualitative study results. This is because trustworthiness and transparency are necessary in a qualitative research study. Thus, teachers' perceptions and narrations have been portrayed, with citations where necessary, in the way described to affect fully the deep meaning of phenomena for reader's understanding.

According to Bryman (2008), reliability of the work done has to be tested in order to ensure the quality and scientific validity for the studies. Reliability here implies to the study being of the same quality even when re-performed. Reliability and stability of the collected data over time and over the situations surrounding the study is necessary for its usefulness and integrity. For the interviews in the study to maintain their authenticity the same study should produce the same results if another person were to do it using the same questionnaires (ibid.). The reliability of a qualitative study as opposed to a quantitative study is that it is very difficult to achieve and therefore, it is possible that the results will differ depending on the interviewees, the settings behind the study, and social environment (Bryman, 2008). This is because phenomenon experienced or perceived by an interviewee may differ from time to time and so differ with a study.

The study's results can point to a possible tendency that may complement another research study in order to create even better awareness. Had the study involved more science teachers in the country then reliability increases as the likelihood of getting even more detailed results would be high and at the same time the results would be the same as the ones we have here. This further implies that external validity will be hard to come by as it is a question of to which extent the study results can be induced and applied to other contexts (Bryman, 2008).

4.6 Data analysis

The way I handled the analysis of the study material is that I replayed them all and started transcribing the material. Thereafter, I started analyzing the interview material by looking for consistent patterns and relevant material in the data. The patterns were coded as a way to be able to process them in the right way. Alvehus (2013) describes coding as a process to find a pattern through material by dissecting and breaking it down in order to build up an analysis. It was a slow and painstaking process but, in the end, the empirical material became structured and easier to prepare for my argumentation in the results of the study. The coding was simple and included a theme name that related to how questions were answered in the study. These coded patterns came to constitute the second- level headings used in the results section.

Two things came to mind as I worked on this study. The interviews were based on the case study questions and as I listened to the interviewee, I jotted down parts of the interview, key words that I found gave extra weight on the answer also noting at which time it was mentioned to the minute and seconds. Examples of extra weighted words were teaching methods, discussions, usage of school textbooks, NTA box, motivation, digital resources, experimental activities, use of outdoors, and understanding of concepts. Finally, after analyzing the empirical data I interpreted it and read through it once more to get a clearer picture of how to continue with the study.

5 RESULTS

The following section represents the results of the empirical study on the science teachers. The teachers' responses are categorized by using subheadings that I perceived relevant during analysis of the interviews. The interviewed teachers will henceforth be known as Teacher 1, (T1), Teacher 2, (T2), Teacher 3, (T3), Teacher 4, (T4) and Teacher 5 (T5).

5.1 Teaching in the classroom

Being a dynamic teaching environment, the classroom is a place where students with different backgrounds and varying personalities and abilities come together every day to attain knowledge in different ways. A teacher is required to utilize innovative and creative teaching methods in order to meet every student's individual needs.

5.1.1 Reference

All the interviewed teachers acknowledged that they used the school curriculum as a planning tool to make sure that they were meeting the necessary knowledge requirements for each of the lessons taught in their various subjects. The use of the textbooks was unanimously accepted as a reference point for lesson planning and were also seen as a means for herding the classes together. This was partly because the teachers followed structured guidelines in their schools for how teaching should both look and be conducted. This was in part the curriculum and lesson plans and in part the local regulations set up within the school for teaching. All the teachers agreed that the guidelines are well founded and work well. Accordingly, some of the teachers thought improvements in this area would also be welcomed as they believed that teaching evolved. Another confirmation from all the teachers was that apart from subject textbooks, they acquired physical teaching materials along with teaching inspiration from so many other places. The most common places were the internet and other literature, but also being out in the nature was something some of the teachers derived teaching inspiration from. According to all the teachers, teaching in the classroom is dynamic and done using many different methods.

5.1.2 Discussions

T1 prefers classroom teaching in combination with other methods like group and class discussions where students work in groups and pairs. The same teacher explained that the attention span of students is short and delicate which meant that it was important to keep in touch with the students to maintain motivation for the lesson. T4 used group and pair work and class discussions afterwards because this was a way of promoting and building an active verbal interaction with students. T5's

lessons usually started with the class going through lesson aims, the purpose of the tasks and the contents of the lesson and vocabulary students needed to be familiar with for the topic. This information was projected in turns on the interactive whiteboards for everyone to see. T5 used tasks from textbooks and worksheets to provide a good ground for class discussions, because here students' difficulties in understanding tasks and questions can easily be dealt with.

T5 commented:

> I want the students to work with the textbooks and topic-related worksheets in class thus working more with factual information. That also provides opportunities for them to understand more about what they learn in English and Swedish subjects in terms of working with factual texts, reading between lines and using strategies to understand texts. Focus on their understanding is total.

By working this way T5 said it helped students acquire knowledge they could use later when in experiments or with science kits (Nature and Technique for All). T4 actually collaborates with the Swedish language teacher where they work together with factual texts and lab reports. Students learn strategies for writing lab reports, factual texts, using vocabularies when writing during their Swedish language classes and this knowledge is then transferred to science lessons. T5 summarizes lessons with class discussions or sometimes pair presentations. The teacher noted that students carry this knowledge with them to the following lesson, which could be hands-on activities. T5 notes that this was like a bridge where knowledge growth flows from the abstract to the more hands-on and practical work, and this cements the knowledge among the students.

T5 described the activities they do in the classroom when working with the topic of buoyancy. The students collect things to test and categorize them according to their characteristics, predict if they will float or sink when immersed in water and go through science concepts connected to the topic. For example, air will increase buoyancy, objects that can sink displace their own volume of water while the floating ones will displace their own weight of water. T1 used a tennis game-like system where a concept was shared to the students and they were to think individually first. They then tossed a tennis ball around in class to everyone to give their thoughts. No one talked without first holding the ball. It was also ok for the students to say "pass". However, one could not use that word more than twice.

T3 argued that it was a challenge to find good strategies to both stimulate and maintain students' focus when teaching science in classroom. Several of the other teachers lamented as well that it was

challenging to meet every student's needs. There were some topics where the variation between those with sufficient pre-knowledge and those without was big. T3 was aware that some students could hide behind others in group discussions or when doing experiments. "Some students for various reasons will hide behind others, let others lead or take the initiative." T3 noted. "Group discussions and peer-to-peer discussions create opportunities for collective learning, however I require individual explanations to questions asked." This was one way to assess their knowledge acquisition in the topic.

T2 first uses discussions in various forms and summarizes with individual work. T2 said the discussions were very welcomed in the lessons. The same teacher explained that class and group discussions were like an arena for sharing knowledge and experiences, where one gave and the other received and vice versa. For example, through discussions the students trained to apply critical thinking when answering questions. Knowledge from other subjects was integrated when students first analyze, then synthesize either individually or in pairs, and lastly, they generalize through discussions. Students are free to share similar experiences and knowledge with the rest of the group or class. These common experiences have significant benefits for their understanding of the subject, concluded T2.

All the teachers use the time in classroom to prepare for experiments the students will do later on. Most of the teachers started with conceptual models, hypotheses or questions that required individual work before working in pairs. This was usually summarized in class discussion. T2 deviated from the norm periodically by giving the students an explanation to a conceptual model and then requiring them to work out how the explanation to the question came about. The students were required to work individually, in groups and then the whole class moved closer to the teacher at the front for a wholesome discussion. T2 believed it was important for the students to understand, not just answering questions. "I know in some situations you cannot have one answer to a question. Because they are all unique, they will have different ways to tackle a model. This brings about healthy discussions." T2 continued to explain that successful discussions required mature social norms in the classroom. Accordingly, T2 believed students must feel confident about expressing themselves in class so the teacher involved the class so much. "Otherwise as the teacher I would be dishing out information and the students just sit there being passive listeners" concluded T2.

All the teachers admitted to introducing subject topics, tasks and also working on after-experiment tasks in the classroom. The teachers also brought outside activities into the classroom, like after field trips. This involved individual documentation and answering of worksheet questions but could also

be pair work followed by group or class discussions. For T4 the class discussions reveal individuals' interests and experiences they have had. The same teacher concludes that it is not possible to connect with all students all the time but some of the time. "It is wise to be realistic and be grateful when you can be able to attract their interest and focus through their experiences."

5.1.3 Students' prior knowledge and experiences

All interviewed teachers agreed that using students' prior knowledge and experience in teaching in order to increase students' interest in science lessons coincided with teachers working on building students' motivation. They also all agreed that being attentive to students' curiosity about aspects in life and actually using that in the lessons, coupled with relevant subject videoclips guaranteed their focus. T1 argued that when students related the lesson to their own experiences or pre-knowledge, then for the teacher, half the work was done.

T1 explained;

> When you succeed as a teacher to link a concept to something students are familiar with, you'll connect their existing knowledge to the new information you are giving them. Students' reality and experiences could come from for example, field trips, their own summer breaks, or their contact with family.

T1 exemplified that parents teaching their children how to sort garbage helped the individual in subject topics like renewable or non-renewable resources because they touched issues like how garbage is disposed of or how to care for our environment. Being realistic, T1 said it was hard to have interesting topics and lessons so they balanced that by varying how they met students' needs in class. T1 believed that teaching science in school will be beneficial if teachers provided rich and authentic experiences of science in the classroom. T1 meant that if students felt comfortable and capable of working on their tasks they would always want to do more. This increased students' motivation on the subject in general as they would want to share with the class of their experience and knowledge. "And who wins by that? Everyone!"

T3 explained that the "aha!" feeling happens momentously when an individual suddenly gets the solution to a problem that they spent a lot of time trying to solve then from relating to existing knowledge or experiences the answer suddenly dawns on them. Using what they have solidifies knowledge acquired and keeps motivation high.

T4 had a similar approach to this method. The teacher explained that being a leader in the class and giving precise, simple and straight forward instructions for the actual tasks should be part of the everyday presentation for science lessons. T4 believed that using down-to-earth examples with the tasks, "adding students' own experiences to it [...], the tasks become easier for the students to tackle and they put effort into the work." T4 meant that the effect of the connection to students' knowledge background was that it would be as a prerequisite for encouragement to increase efforts in tackling tasks in class.

T5 coupled discussions with students' own experiences and admitted to being overly generous to letting the students relate their experiences to the lesson topic. The same teacher believed that this action created a bond of trust and comfort for the students and also solidified knowledge acquisition. Throughout the interview the teachers acknowledged the importance of knowing students' interests and experiences as a gateway to creating good relationships. On the effects of using students' pre-knowledge and experiences, T4 noted that it was easier to relate to the individual when the teacher knew about students' interests and experiences outside of school.

T4 explains:

> Someone in my class loves fishing. They got the chance to teach the whole class about fish habitats in the region. Even I learned from that experience! The student was so happy from the experience and it showed from the work he handed in that he loved the lesson.

Consequently, T4 explained that as a teacher they had to work with different teaching strategies and methods. Factual information could for example be linked to an experiment and class discussions to an oncoming outdoor activity. T4 believed that varying teaching in this way helped the students understand the subject in general.

5.1.4 Repetition

All the interviewed teachers used repetition in differing ways when teaching. They used repetition after experiments or laboratory tasks in order to help the students to remember and gain good understanding for the work they did. T1 for example, usually started from where the lesson stopped previously so she could "collect everyone from their point of understanding to bring them to the lesson aim of the day." T5 periodically used exit notes with specific questions to reinforce remembrance of what was learned and subsequently to know what the students had learned at that point. (Exit notes are usually small pieces of paper with questions or a concept written or drawn on

it for the students to answer or solve towards the end of a lesson). T5 explained that the students handed in their exit notes to the teacher before the lesson ended. All the notes were read and evaluated to assess the students' understanding. It also gave a broader insight into how the next lesson or homework would be planned. T5 noted;

> Students frequently use repetition through exit notes, sometimes I do it orally where everyone gets to say something about the area we have touched, or they write down their answers and explanations.

According to T4, repetition was a good method for learning and preferable to summarize what was learned because the individual retells what was done or said but by using their own formulations to describe their new knowledge. T2 practiced repetition through quizzes that required students to describe what they learned or how they got a certain result from the experiment or a laboratory task. T2 explained that with every lesson, there was no guarantee that all students understood and remembered what was taught. Lesson time management caused dilemmas between moving on to the next topic or taking more time to maximize learning of the actual task. T2 explains, "Students can forget what they learned, even when they seem to have understood the concepts you are teaching." T2 noted that repetition becomes important because it deepens and enhances the learning process.

T2 went on to explain that it was important that the students construct their own understanding regardless of what the teacher taught or what they read from the text books by explaining or describing with their own words. Therefore, T2 used repetition to reinforce knowledge and understanding of class tasks. Both T2 and T1 believed that repetition was of primary importance for the students' actual learning. It focuses on the individual and so it's a very important method for enhancing learning.

T1 explained that the greater picture for the teacher was to see everything with a bird's eye view:

> You are doing repetitions of a moment in a lesson to capture knowledge so the students can remember and understand. Then you don't end up re-teaching the whole lesson contents again. No one has time for that. As a teacher then, I try to invest my time in this process.

T3 explained about the problem posed by texts in science books. The language in the science texts books could be unfamiliar to many students and was permeated with sometimes difficult vocabulary.

The basic repetition method of reading vocabulary and concepts out aloud to students who listened and then repeated helped in many ways. The students with Swedish as a second language had a greater need for this learning method as it was important for language acquisition. By repeating after the teacher, the students processed and stored the vocabulary and could use it later on, especially in hands-on activities or in tests. No other teacher interviewed used repeating in this way as much as T3 did. By using repetition T5 acknowledged that the children are forced to think through the vocabulary which enhances their knowledge. T5 preferred using repetition through homework where students memorized vocabulary.

5.1.5 ICT[1]- enabled learning

T1, T2 and T5 used laptops and T3 and T4 used tablets to optimize their teaching. All the teachers in the study believed that ICT-enabled learning was a positive thing in school and described very positive experiences from using it in teaching, although with notable variations. T1 and T2 used ICT-enabled learning constantly for both theory and practical lessons. T3 and T4 noted that they could use technology even when out in the nature by taking pictures and for beginning the group activities that students would work on together. These teachers acknowledged that ICT made their work easier and students worked faster. Almost all the homework was posted and shared to students using technology. This option was not available for T5 who's students had limited access to digital tools and for this reason focused on the use of schoolbooks and textbooks. However, they were free to complement knowledge with digital tools.

With interactive whiteboards, all the teachers easily accessed and used a variety of media tools and applications during science lessons. From the teacher's laptop or tablet, different presentations could be displayed on the interactive whiteboards, often interspersed with concepts. This enabled learning to occur in a comfortable way, explained T3. Another thing the teachers mentioned with the use of interactive whiteboards was that it allowed them to be very flexible with their teaching in general. An example was that when T3 noticed a lack of focus in the classroom, "I usually give them a short break and students chose to dance to a song." "We used my laptop to do that."

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1 *I define Information and Communication Technology (ICT) in education loosely as modes of optimizing teaching and learning by using varied technology.*

To counter “dull topic lessons” (as T1 called them) where students worked on textbooks and worksheets only, T1 used visualization both in classroom teaching and in laboratory sessions. The interactive whiteboards were used to display lesson material like video or audio clips and photos. T3 optimized teaching with the interactive whiteboards by using it with tablets for information search and retrieval together with students. “They do it either individually or in pairs. Students search for the answers to follow-up questions to get the most out of the activity.”

Because T1 started lessons from where the class previously finished, ICT made it easier to bring out information from previous lessons. It was easier for T1 to make things clear with the interactive whiteboards compared with the regular whiteboards. On a certain day of the week, T3’s classes were divided into several small groups for alternating lessons between science and two other subjects. A photo of a students’ finished configuration table from the first group was used to explain to the next group about the expected procedures for the experiment tasks.

The teachers' general reaction to using ICT-enabled learning was that it contributed to teaching in a positive way. The teachers agreed that they experienced more engagement from the students. One way to enhance student attentiveness was to use interactive technology as a motivational tool. T1 and T5 concluded that they noticed over time that students were motivated and engaged with the lesson partly because it was easier to get involved with them when they could see a presentation or vocabulary they needed to use with their tasks. They were more likely to remember what they learned and keep the knowledge partly because they saw and used these words, concepts or vocabulary. T5 argued that writing the aim of the lesson and topic on the interactive whiteboards was fundamental to getting a good start with lessons. It was beneficial to students when difficult or new vocabulary were written or projected on the interactive whiteboards for the students. T5 projected lesson aims on the interactive whiteboards during the lesson:

> I find it easy when I can pause in the middle of the lesson. For example, when a student has explained their answer in a certain way, I will remind the class of what the student has just said and point to the interactive whiteboard so that they can see and connect the lesson aims to the answer.

According to T3 students with Swedish as a second or even third language benefited from using the interactive whiteboards in that projected information helped them to see how the words and phrases are arranged and spelled. T4 explained that technology enabled interactions with students on a whole new level compared with before as it facilitated learning both in and out of classrooms. The

students had access to lesson briefs, homework and other tasks electronically. They also had access to their assessments online. “This would take more time and work to do manually,” T4 concluded.

5.2 Investigative working methods

In this section I have decided to have both laboratory work and experiments under the same category because both use investigative structures and themes to meet educational objectives.

5.2.1 Laboratory and experiments

Concerning laboratory work, teachers had differing facilities that may be due to economic reasons or lack of space. T1 and T2 had access to laboratories at their schools. Both teachers also accounted for a general feeling of excitement by the students when it was time for laboratory classes. More often than not the lessons worked in appointed groups or in pairs. Both teachers share the same view about laboratory work in that they saw it as a fundamental part of science teaching and learning. All the teachers acknowledged that laboratory work (T1 and T2) and experiments (all interviewed teachers), were better when structured goals of the session were conveyed to the students, especially written on the whiteboard. Moreover, the teachers stressed the importance of outlining laboratory and experiment tasks. T5 mentioned that explaining this more than once was important to avoid mishaps later on.

T1 explained the laboratory lessons helped students to apply learned concepts to new experiences by exploring their understanding and witnessing their presumptions proved right or wrong. This could be through investigating elements in chemistry, sorting plants in biology, working on a programming code in physics or technology, etc. According to the same teacher it was important to explain to the students the pedagogical purpose of the tasks in the lesson and aims of the lesson.

> I let them know in the previous lesson about the purpose of the coming laboratory lesson and how the lesson fits in the planned learning process. “It is not hullabaloo!”

With laboratory experiences students were able to deepen their understanding of theoretical concepts and this enhanced their understanding, noted T1. T3, T4 and T5, all agreed that the absence of a laboratory did not stop them from mixing their theory lessons with experiments. T3 argued that by working with experiments students could clarify hypotheses and try out their perceptions. According to T3 this experience stimulated students’ understanding of different phenomena while developing critical thinking. After experiments the students usually deepened knowledge gained from experiments by doing assignments like a lab documentation individually.

For T5 doing experiments made every aspect of learning active as students worked together in groups to solve a problem, test a concept after predicting a hypothesis and compare their findings with others. T5 believed that students developed their data analysis and collaborative skills. T3 had a similar view in that it created a sense of cooperative learning that lifted everyone in the class. In general, all the teachers did see science laboratory lessons and experiments as unique learning arenas where students worked cooperatively in small groups to apply scientific phenomena.

T4 realized that students learn in many different ways and therefore varied theory and experiments to give the student a possibility to learn in the best way possible, by capturing their interest. T4 had access to NTA (Nature and Technology for All) boxes. These boxes had digital material, tutorials and laboratory material adapted to specific topics of science subjects. T3 and T4 discovered that using the boxes with specific themes gave a progression in which experimental work was planned into smaller activities that together created greater understanding. Students could therefore compare their work and come to general conclusions as they finished the subject area. According to these teachers the progression in these experiments coincided with educational goals of the subject area.

All the teachers agreed that experimental activities should be part of the learning process to help enhance the students' learning process. To be effective they had to be coupled with other teaching methods. T2 and T3 emphasized that they always try to give the students a theoretical basis before extending it to practical activity. T1, explained; "When you combine theory and practical activity you attract students at various places, some with the practical activity, and some with the theory."

5.3 Outdoor learning

Of the teachers interviewed only T2 was not delighted with outdoor learning activities with the reason being "discomfort in wandering about." Although T2 did mention that the science classes did venture out into the nature "sometimes during the course of the term." The other teachers reported that they had many science trips and excursions and that students always showed more focus on the post-trip lessons. T3 suggested that maybe being out in nature for science lessons helped reduce stress and energized the students.

T5 was excited when explaining;

> Because we can do this, the students' interest in the subject increases and this is a big factor in wanting to learn. The fact that the students can move about and explore the surroundings, see phenomena in context, like where insects live, what polluted water looks like etc., can encourage students to make interconnections with their surroundings.

> Using their senses in practice when outdoors can also help the students to understand the need to see their world ecologically.

T3 noted that the fact that the students could see, touch, taste and understand the effects the environment had on living things or how we interacted with each other, this drew parallels with learning for the students. T3 explained that using students' pre-knowledge when teaching in these situations provided opportunities to capture their interest and attention, which is very challenging. T3 and T5 loved to be outside with their classes for science lessons. T3 explained that it is possible because of the proximity to the forests and nature reserves. This explanation was also shared by T5.

T4 mentioned that outside activities allowed for good science work like understanding how simple science investigations were commonly done. T4 gave an example where the students were organized in groups to conduct investigations on insects' habitats in the local forest. Each group carried out the investigation using the same rules and procedures. By working on the habitats, the students could create a picture from the data collected that showed different processes in a habitat. Later on, the students converged together to share their results in a discussion in the classroom with pictures and documentations already done on the iPads. The classroom session was for the students to asses if they worked according to the lesson concept. They also evaluated their conclusions from the data the groups collected in the forest. T4 reported that this activity was deemed fun and positive by the students and that benefits included individuals in groups having more input on how to conduct the investigations. Students also helped each other and discussed issues that arose. "This way the students could also learn from each other and have fun together.

6 DISCUSSION

6.1 Method discussion

According to Bryman (2008) a qualitative study method is suitable for this empirical study because it aims to perceive interviewed teachers' experiences, which is what was done in this study. This study comprises of semi structured interviews from 5 active teachers that teach science, mathematics and technology in the elementary school grades between 4 and 7. The study was initiated through mail contact with prospective teachers. Initially the plan was to have eight teachers spread across the province of Jönköping. However, of the eight requested, only 5 accepted and provided feedback on how and when they could do the interviews. For the purposes of having a relevant picture of my study, I chose to work only with legitimate science teachers. Three of the teachers have a license to teach both in the middle school and the upper primary. This would be grades four to grades nine in the primary schools in Sweden. Two of the teachers have additional international teaching backgrounds and several years of experience in teaching both the middle school and the upper primary school both in the U.K. and U.S.A. The group was rich in quality. According to Bryman (2008) this also increased the probability of acquiring relevant insight and knowledge for the study. Likewise, Denscombe (2016) also describes that selection of interviewees is a conscious decision based on the information, their knowledge, and how relevant it is to the study.

Prior to the interviews, relevant literature was examined to be able to have a clear picture of how to follow interview study protocol, thereby literature from Vetenskapsrådet (2017) among others. Preparatory scrutiny of the literature gave good grounds of guidance on what to think and do with the chosen empirical method. The choice to use semi structured interview questionnaires was to allow for openness from the respondent to formulate their answers in a way best understood for them (Bryman, 2008). To this was the option to have follow up questions if something wasn't clear, also part of the reason for using semi structured interviews. These follow up questions were impromptu and served to moor on a broader understanding to reduce the impact of personal interpretation Stukát (2011). Interviewees described their perceptions of experiences and methods they used to teach their subjects in a way that could help me register and interpret phenomena. In this way, I could correlate to the aims and questions the empirical data set out to answer. The questionnaire used in the interviews with the interviewees resulted in rich and collaborative exchanges. I see this as a strong side of the interview, partly because the questionnaire was designed with an openness for follow-up questions. The provision of questionnaires in this empirical study should wholly emphasize its credibility, as provided in Appendix 2.

The interviews were all done on the interviewees' terms in that they chose the time and day to be interviewed. I offered assurance of space serenity and secrecy in handling the personal information through anonymizing (Eklund 2012; Kvale & Brinkman, 2014). To counter the possibility of having to do interviews in unconducive areas it was also decided to have them after school hours. According to Eklund (2012) the interview should take place in a comfortable atmosphere for the interviewees.

The interviews were also recorded to undercut any loss of information that might have been crucial to the empirical study. This is in accordance with what is described by Bryman (2008) in that recording is not necessarily of any importance, however it helps reduce the risk of losing study data. Additionally, within a qualitative empiric study it is imperative to transcribe the collected data and thematize it. Personalities, experiences and assumed rolls by the teachers and their immediate environments, may have laid the ground for differing perceptions that became evident throughout the transcribing process. According to Bryman (2008) this is a risk possibility in the study. Furthermore, the jotting down of things perceived to be important and notation of where in the interviews this was said accounted for the rich data collected. For example, at 10:40 subject talks about their facial expressions and things they say that students see and know (Denscombe, 2016). This helped in thematizing the data later when transcribing it as opposed to solely using paper and pen during the interviews.

Editing of the interview data included thematic analysis to summarize collected data into desirable categories. Based on the scale of the interviews, it was deemed easiest to use (Bryman, 2008). A crucial thing to think about here is the problem with deciphering data in a way that correlates with what the interviewees described. The presence of recordings in interviews is therefore crucial to circumnavigate such reliability issues. Lastly, qualitative research data can be difficult to generalize mainly because there are many parameters to consider, like interviewees' differing surroundings and workplaces (Bryman, 2008). Though for this study group we can with a certain degree of certainty generalize that the data given is in line with given criteria and its validity and reliability is sound.

6.2 Result discussion

Based on the results, certain tendencies could be distinguished, such as the teachers' love for science. Their descriptions of approaches to learning science gave opportunities to enhance students' experiences to enable them to achieve desirable scientific literacy. They also challenged the students to be creative through various balanced working methods. This conforms with the Swedish curriculum for teaching in primary schools (Skolverket, 2019).

6.2.1 Teaching in the classroom

School curriculum was widely accepted as a reference guide when using subject literature for lesson planning. Something highlighted by all the teachers in the study was the importance of routines and clear-cut information shared when it came to working with textbooks. The underlying reason was to reduce potential cognitive complexity brought about by students not understanding and thus not following instructions. To typify this, Andersson et al., (2005) describes that the teacher's role in learning science is crucial to the students' motivation for the subject. The emphasis is that an ongoing interaction with the teacher is significantly better for learning than just working with books and other materials. This is in conjunction with Vygotsky's theory in that we learn first through person-to-person interactions and then individually through an internalization process that leads to deep understanding (Blake & Pope, 2008).

Sindhu (2014) noted that text books are important teaching and learning materials that usually prescribed and dictated teachers' pedagogical practices. Furthermore, textbooks play an important role in influencing teachers' choice of content and pedagogy (ibid., 2014). The fact that all interviewees' schools followed structured guidelines for teaching science subjects solidifies all the more Sindhu's research findings. To overcome the tyranny of having to rely on just the school provided textbooks and thus easing the pressure of having to work in a standardized way, the teachers acquired teaching material and inspiration from so many other places (ibid., 2014). Usage of the internet as a resource to teaching material has grown over time and found a place in the classroom. This means that teachers must take initiative to delve into the plethora of learning resources found in the world wide web (ibid., 2014). The study has shown that the use of text books overlaps with the use of other resources especially in pedagogical planning. Research has shown that teaching science can be successful and learning rich when a teacher is motivated to go beyond the textbook and beyond the focus of topic related engagement. This makes classroom learning lively and engaging (Sindhu, 2014).

Something in this study that proved to be of great importance in lessons was a teacher's use of discussions. The use of class discussions varied from pairs to small groups to whole class. Among the reasons for shifting between different group settings in a lesson was counter focus deficit among students. All five teachers explained that the best way to have the attention of the students in science lessons was to vary the activities in class during lessons by shifting from teacher-led to peer to group and class discussions. The coaction this method creates can be likened to Vygotsky, who emphasized the roles of social interaction and instruction (Blake & Pope, 2008). For example, T4 paired students to work on an activity after which a class discussion would ensue before students did individual

work. This, according to the teacher, promoted an active verbal interaction between students and the teacher.

T2 described discussions as arenas for sharing knowledge and experiences where an aspect of give and take among the students is created when students engage in these kinds of learning methods. Teachers' ability to motivate and inspire students to partake in classroom discussions has enormous value in maintaining motivation for the subject. Working on student participation and achievement in classroom discussions through the teacher's leadership is synergetic to Vygotsky's ZPD. This is where students require leadership from the teacher when conducting discussions and conversations that herd students towards subject and lesson aims (Blake & Pope, 2008). In this way, teachers help the students construct their own knowledge through interaction with other students and accomplish what is required of them through the instruction and leadership of the teacher.

During the interview T1 shared that the class used a tennis game-like system where a concept was shared to the class and they were to think individually at first. Afterwards, the class would resort to a discussion whereby the tennis ball was tossed about between students and the one who caught the ball shared their thoughts. This interaction between students and the teacher is what Säljö (2015) describes as knowledge coming into play and being acquired through interaction between people as they coordinate their perspectives collectively. Hence the origin of knowledge is in the interaction. Subsequently, (Säljö, 2015) indicated that knowledge can thus be understood as something that arises and gains meaning in the interaction between these students. It is fair to conclude then that T3 also relates to the above argument by Säljö that group and peer-to-peer discussions as a teaching method creates opportunities for both collective and individual knowledge acquisition.

Earlier research supports science teachers' use of students' pre-knowledge and experiences in their lesson. According to Lundin (2007) science teachers can use students' experiences in various ways. They can use them as a means to implement a lesson or as part of a subject topic content. T1 elucidated that when students related the lesson to their own experiences or pre-knowledge, then for the teacher, half the work was done. In agreement Sjøberg (2005) noted that it was important for teachers to know students' interests and use them in their teaching, while Hattie (2014) stresses the importance of taking advantage of students' interests, noting that coupling them with what is learned in science could arouse a curiosity and thirst to learn more. In this way, the teacher could help the students to bridge the gap between science and their individual experiences. In the study it was evident that all the teachers claimed to use students' experiences and pre-knowledge in teaching.

The teachers believed that when teachers succeed in linking concepts to students' experiences, then they connect students' existing knowledge to the new information they received. It does matter then, how the material is presented to the class. As Sjøberg (2005) explicates thus; "How should teaching material be presented in a way that promotes students' learning? Because it is learning that is the goal, and what a student learns, is not always the same as the one teacher teaches, or what is presented in a textbook." (p. 33). In this context, the results have indicated that classroom teaching almost always has moments where pre-knowledge and experience come to play. For T4 the effect of the connection to students' knowledge background was a prerequisite for encouragement to increase efforts in tackling tasks in class. From the study, it is evident that teachers' approaches to teaching science lessons tend to demonstrate the value of creating a deep sense of experience and understanding for the students in order to meet curriculum goals. This is hereby begun through the unpacking of individuals' pre-knowledge and prior experiences. The results of the empirical study also indicate that teachers work consciously with variation of teaching methods to capture students' motivation and provide common experiences in class. This is a way of highlighting their capacities in science lessons which is good for their motivation.

All the interviewed teachers' recognition that repetition was a powerful and significant way to improve performance in class was indicated in the study. By analyzing the cognitive view of T4 it is established that repetition was a good method for learning and preferable to summarize what was learned because the individual retells what was done or said using their own formulations to describe their new knowledge. For example, T2's view that repetition helped the students construct their own understanding regardless of what the teacher taught, embodies Piaget's view that emphasizes that an individual learns when they construct their own conceptual scheme (Blake & Pope, 2008). A key point by Blake and Pope (2008) is that Piaget's view is beneficial in the classroom in that it helps teachers develop a better understanding of how students are thinking. It also enables teachers to align their teaching methods with students' cognitive levels by helping them construct knowledge (ibid.). By utilizing Vygotsky's ZPD, whereby students are given opportunities to put to practice what learned under teachers' guidance, thereby building and internalizing knowledge. To summarize then, results show that the students' embodiment of the conceptualization of what is being taught in science is their knowledge acquisition.

From the study it is possible to say that quantity gives quality in that the extensive use of ICT-enabled teaching has been shown to increase students' motivation and interest in learning as acclaimed by the teachers. According to Hylén (2013) increased motivation for schoolwork and student engagement in learning was achieved with the introduction of digitalized teaching methods

when used in combination with other methods. Interviewed teachers in this study explained that they varied ICT-enabled teaching methods as this contributed to learning in a positive way. The teachers also agreed that they experienced more engagement from the students. A study conducted by Uluyol and Sahin (2016) concluded that teachers noticed an increase in students' knowledge and skills, such as communication, problem solving, understanding and analysis through ICT- enabled learning (ibid.). Earlier research by Murcia and Sheffield (2010) indicated that ICT- enabled teaching contributed to students' motivation and engagement to explore the role of science in understanding their surroundings and scientific concepts. Research has indicated that the teacher factor is of greater importance than the technology itself when it comes to optimizing ICT in teaching, in that enhancing students' conceptual competence, motivation and even performance results is an effect of the teachers rather than ICT in itself (De-Witte & Rogge, 2014). Notwithstanding the above results, research has also shown that provision of ICT-enabled classroom teaching does not really guarantee enhanced educational performance by the students. It is instead more important how it is used (Livingstone, 2012). Even though ICT-enabled learning has had the expectation to enhance variable knowledge acquisition in classrooms for students, it also tends to alienate teachers (ibid.). Teachers are often left to fit technology in their pedagogical preparations as best they could to suit teaching and students' individual needs, among other variables (Livingstone, 2012).

The dynamic of technology use in teaching science has changed how a teacher interacts with students during lessons. The concept of interaction as per Vygotsky has shifted as students prove to be more and more independent in the way they seek knowledge. One of the negative sides is that what is sought could be disinformation.

6.2.2 Investigative working methods

According to the results, the teachers seemed convinced that there was an enthusiasm from the students in wanting to participate in laboratory activities and class experiments. For example, both T1 and T2 noted the excitement that came with laboratory activities and experiential work in class. T1 explained that the laboratory lessons' aim was to help students apply learned concepts to new experiences by exploring their understandings and witnessing their presumptions proved right or wrong. This could be through investigating elements in chemistry, sorting plants in biology, working on a programming code in physics or technology, etc. Cimer et al. (2007) argued that students need to test and develop their models and thought processes in familiar context, in order to experience relevant learning. This brings us closer to Piaget's view on exploring to acquire new knowledge or ascertain the actual one. All teachers unanimously acknowledged that investigative working methods gave better results than when structured goals were conveyed to the students. Successful

practical activities depend on how teachers present them to the students in their science lessons as this can have significant consequences on whether the lesson aims and purposes are achieved (Millar, 2010).

When teachers use effective instructional methods and strategies based on the developmental and cognitive psychology theories of Jean Piaget and Lev Vygotsky, they facilitate opportunities for students to acquire knowledge (Blake & Pope, 2008). T5 clarified that experiments made every aspect of learning active as students worked together in groups to solve a problem, test a concept after predicting a hypothesis, and compare their findings with others. This can be aligned to Dewey's perspective in that when teachers work in this way they stimulate students' curiosity from the energetic explorations of physical curiosity, via the questioning and investigations, to the more sustained and systematic problem-solving of intellectual curiosity (Luff, 2018). Dewey described that experience is an interaction between doing something and seeing what it leads to. The individual learns when they understand the connection between the activity and the result (Imsen, 2006). The teachers had a similar view in that these learning methods created a sense of cooperative learning that lifted everyone in the class. In general, all the teachers did see science laboratory lessons and experiments as unique opportunities where students worked cooperatively in small groups to explain scientific phenomena. Dewey believed that curiosity in students can be stimulated by processes of enquiry and experiments (Luff, 2018). NTA boxes used in science lessons by the teachers interviewed facilitated progression towards curriculum goals through the process of finding meaning, thinking independently, and questioning of the social world. One could analogize these approaches to Vygotsky's theory of learning which delineates the individual's learning process through communication with others as a prominent aspect of development. The interaction between induction and deduction in teaching science was, according to the teachers, a way to creatively challenge everyone in class.

6.2.3 Outdoor learning

From the study result it can be understood the teachers shared an interest for nature and outdoor activities. These teachers also acknowledged that having students out in the nature helped in their cognitive development. This understanding is shared by Blom and Fägerstam (2013) whose research on learning biology outdoors shed light on outdoor learning activities by describing that students are offered learning in firsthand, reality-based contexts and this could facilitate deeper understanding of subject content. Compared to indoor lessons, outdoor science lessons were more student-centered filled with different contexts, hands-on activities, and content knowledge that formed relevant parts of biology teaching and learning (ibid., 2013). This is evident in T5's explanation about students'

evident interest in the subject growing over time. According to T5 outdoor science lessons enables students to make interconnections with their surroundings and help them understand their world with a deeper meaning. Findings from research done by Hofstein and Rosenfeld (1996) indicated that outdoor activities provided some the most valuable informal science learning for students. Outdoor learning also facilitated long-lasting effects on students' cognitive abilities in that students experienced a richer contextualized, hands-on science education as active participants (Blom & Fägerstam 2013).

Outdoor learning environments allowed students to interact "physically and intellectually with instructional materials through hands-on experimentation and minds-on reflection" which made huge impacts on students' learning of science (Hofstein & Rosenfeld 1996). Earlier mentioned was the belief from John Dewey that children should be provided with necessary tools and skills to promote their individual growth and development (Erwin & Kipness, 1997). By conducting outdoor lessons teachers advocate for John Dewey's view that acknowledged the need for children to be active participants in surroundings that stimulate their imagination, exploration, and interpersonal interactions. Dewey understood that nature was not external to human experience, rather humans were within and part of nature (Luff, 2018).

According to Dewey, individuals learn from experiences they gather by doing things, not by being influenced by external stimulation. Dewey felt that experience is an interaction between doing something and seeing what it leads to (Imsen, 2006). Experiential learning was a process of meaning-making that was vital for students. Active exploration of their surroundings facilitated progressive knowledge construction within specific social, cultural and ecological contexts (Luff, 2018). That the students functioned well in groups, gave more input on topics, and helped each other, are but a few of the benefits of outdoor science learning. These shared experiences functioned as a foundation for continued learning in the classroom. Indeed then, coupling outdoor and indoor learning activities helped diversify the repertoire of learning opportunities among students (Hofstein & Rosenfeld, 1996).

6.3 Conclusion and future research issues

A review of previous research (Rudolph, 2008) has identified a range of methods a science teacher can use in their lessons in order to motivate students in elementary schools. This study has gone further to explore and describe phenomena as is described by the teachers in their work. The presentation of the questions as a reference point and results of the study, while not extensively conducted, could give interesting insight for future studies.

The findings leave little room for doubt that teachers work to promote understanding in their lessons by relating to students' interests, experiences, and values. By using integrated working methods that bring about tangible results the teachers methods created a conducive environment for both stimulation and a greater academic commitment among students. To achieve positive student engagement in lessons, it seems then that teachers need to provide a variety of learning opportunities. Teachers should scaffold through modeling, provide for collaboration etc., and encourage thinking and understanding in their science lessons (Luff, 2018).

As per the results of this study, it is perceived that the interviewed teachers base their teaching on the previously described learning theories. As they are a vital part of our education and learning in Sweden it is fair to point out that the integrating of these teaching and learning theories fit different scenarios of learning differently. Therefore, they absolutely need not necessarily be considered as contradicting each other as much as complementing the other as evidently shown in the study. Because learning theories have differing focus in learning, teachers would be well placed in their pedagogical aspirations by using different learning perspectives to be able to cater for different aspects of learning. Furthermore, different curriculum needs may require different theoretical learning focus. The teachers' descriptions about their teaching methods suggest that they seek theoretical support and inspiration from various learning theories, more so, in this study, from sociocultural perspective and through pragmatism.

This study has also revealed that the teachers make use of an accumulated repertoire of subject know-how and are actively using it in various ways to impact student learning and motivation. As teachers, the right attitude would be to understand which methods fit the class dynamic as this will affect student engagement and subsequently, performance and motivation. It is clear therefore, that the need to adapt teaching to cater for students' individual needs should be deemed a good reason to help unlock students' potential in their quest for knowledge acquisition.

For further studies, it would be interesting to study by both interviews and observations the methods discussed here to understand how students perceive they learn science subjects and if these methods actually enhance their conceptual competence and motivation in the subjects. It would also be interesting to look more specifically into whether these methods are suitable for students with special needs. Also, whether if in a normal environment the results can be generalized to other science teachers in the same contexts.

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Letter of consent

Appendix 1

Hej!

Jag studerar till grundlärare med inriktning mot arbete inom årkurs 4 till 6 och ska nu skriva mitt examensarbete. Temat för mitt examensarbete är lärares undervisningsmetoder inom NO. Syftet med studien är att undersöka vilka undervisningsmetoder NO lärare anser att de använder när de undervisar i naturorienterande ämnen och om dessa metoder kan motivera eleverna i deras lärande.

Jag skulle vara tacksam om du kunde tänka dig att vara med i studien, där jag kommer att göra en intervju med er genom att spela in den.

Allt inspelat material kommer att avidentifieras. Materialet kommer bara att användas för forskning och inget annat.

Deltagandet är frivilligt och du har rätt att avbryta ditt deltagande när som helst och utan att ange någon anledning.

Om du har frågor om studien, hör gärna av dig till mig.

Hälsningar

Rudolph Ndurya.

Kontaktuppgifter

Rudolph S. Ndurya ndru15Wt@studnet.ju.se

Genom att skriva under här nedanför intygar du att du tagit del av informationen.

Accepterar du att delta i studien?

☐ *Ja, jag har tagit del av ovanstående information och accepterar att delta i studien.*

Om du deltar i studien, godkänner du att dina personuppgifter behandlas för utbildningsändamål i enlighet med ovanstående information?

☐ *Ja, jag har tagit del av ovanstående information och godkänner att mina personuppgifter behandlas för utbildningsändamål i enlighet med ovanstående information.*

Om du deltar i studien, godkänner du att dina personuppgifter behandlas för forskningsändamål i enlighet med ovanstående information?

☐ *Ja, jag har tagit del av ovanstående information och godkänner att mina personuppgifter behandlas för forskningsändamål i enlighet med ovanstående information.*

Namnteckning: Namnförtydligande:

_____ _____

Interview questionnaire

Appendix 2

- Hur skulle ni beskriva er själv och er vardag på skolan?
  Följdfrågan: Vad är det som driver er?

- Vilken utbildning/behörighet har du?

- Hur utvecklar ni er undervisning för att öka elevernas förståelse? Hur presenteras NO-undervisningen för att väcka elevernas intresse och motivation till ämnet?

- Vilka undervisningsmetoder anser ni att ni använder när ni undervisar inom NO och av vilken särskild anledning har ni valt just dessa metoder?

- När ni undervisar inom NO använder ni olika metoder för teori och de olika praktiskt arbeten? Om ja, vilka är några av dem och hur beskriver ni dem?

- Beskriv en typisk lektion där ni arbetar teoretiskt, och med experiment eller laboration.

- Vad anser ni som NO lärare är allra viktigast att arbeta med vid NO undervisningen? (ämnesmässigt- kemi, biologi, fysik; samt teknik)

- Följer er skola bestämda riktlinjer för hur ni ska hålla undervisning inom NO? Om detta stämmer, anser ni att dessa riktlinjerna fungerar bra? Om inte, vad tror ni behöver göras för att göra det bättre?

- Om ni fick ändra på något i undervisningen, vad skulle det vara då? Varför just detta?

- När vet ni att er undervisning har lyckats/inte lyckts? Vad gör du då?

- Hur ser du på läromedlets betydelse för undervisningen? – vilka för och nackdelar ser du?

- Vilken betydelse har motivationen för lärandet inom NO? följdfråga -Vilka förklaringar ser du i elevernas brist på motivation

My guidelines for the interview process

Appendix 3

Planning of my interview:

- Always have two phones and they should be set on flight mode. Nothing should spoil or interfere with the process.
- You don't own the interview, the respondent does. You ask, they describe, you record and note down. If possible, ask the question you need an answer to.
- Let yourself be flexible. If the respondent wants to add information, let them. Use, describe further, please continue, affirm and be as good a listener as possible.
- Remember the respondent owns the interview, give them leeway so they are comfortable with the process. If they will start with the last question, fine. Let them.
- Encourage quiet and comfortable interview places.

When i start the interview:

- Record the conversation and have back-up phone at ready.
- Self-introduction
- Explain reason for doing interview but keep the thesis aims away from the respondent.
- Explain about the privacy measures being taken in the course of working with the empirical study.
- Estimated time length of the interview.
- Should not be opinionated with respondent.
- Accept they may want time to think and maybe pause in their descriptions.

After the interview:

- Do they have anything else to add to the interview even if it is not raised as a question?
- Thank them for participating
- Again, remind them of the privacy measures you are taking with the study.
- Stop the recording